

FileEditViewCompileSimulateAddSourceToolsLayoutBookmarksWindowHelp

ColumnLayoutDefault

100 ps

LayoutSimulate

Search:

Documents/ModelSim/PROJECT/fulladder Gate level/Fulladder

Objects

Users/Ragavan/Downloads/Documents/ModelSim/PROJECT/fulladder Gate level/ArithmeticLogicUnit_8bit.sv (/ArithmeticLogicUnit_8bit_tb/DUT) - Default

Name	Status	Type	Order	M
multiplier_tb.sv	✓	SystemVerilog	8	0t
ripple_carryadder.sv	✓	SystemVerilog	3	0t
fulladder_gatelevel_tb.sv	✓	SystemVerilog	2	0t
rcadder_16bit.sv	✓	SystemVerilog	7	0t
multiplier_8x8_gatelevel....	✓	SystemVerilog	6	0t
ArithmeticLogicUnit_8bit.s...	✓	SystemVerilog	4	0t
ArithmeticLogicUnit_8bit_...	✓	SystemVerilog	5	0t
fulladder_gatelevel.sv	✓	SystemVerilog	0	0t
halfadder_gatelevel.sv	✓	SystemVerilog	1	0t

Name	Value	Kind	5 ps
a	0000...	Pack...	Internal
b	0000...	Pack...	Internal
sel	00	Pack...	Internal
result	0000...	Net	Internal

Processes (Hierarchy)

Name	Type (filtered)	State
#ALWAYS#42	Always	Wait
#INITIAL#15	Initial	Active

LibraryProjectsim

Transcript

Ln#

```
1 module ArithmeticLogicUnit_8bit (
2     input [7:0] a,
3     input [7:0] b,
4     input [1:0] sel,
5     output reg [15:0] result
6 );
7
8 wire [7:0] add_result;
9 wire [7:0] sub_result;
10 wire [15:0] mult_result;
11 wire add_cout;
12 wire sub_cout;
13 wire [7:0] b_comp;
14
15 not (b_comp[0], b[0]);
16 not (b_comp[1], b[1]);
17 not (b_comp[2], b[2]);
18 not (b_comp[3], b[3]);
19 not (b_comp[4], b[4]);
20 not (b_comp[5], b[5]);
21 not (b_comp[6], b[6]);
22 not (b_comp[7], b[7]);
23
24 rcadder_8bit ADD(
25     .a(a),
26     .b(b),
27     .cin(1'b0),
28     .sum(add_result),
29     .cout(add_cout)
30 );
31
32 rcadder_8bit SUB(
33     .a(a),
34     .b(b_comp),
35     .cin(1'b1),
36     .sum(sub_result),
37     .cout(sub_cout)
38 );
39
40 multiplier_8x8_gatelevel MUL(
41     .a(a),
42     .b(b),
43     .product(mult_result)
44 );
45
46
47
48 always @(*)
```

Ln: 32 Col: 17

**

Project : Fulladder

Now: 80 ps Delta: 0

sim:/ArithmeticLogicUnit_8bit_tb/#INITIAL#15

FileEditViewCompileSimulateAddSourceToolsLayoutBookmarksWindowHelp

ColumnLayoutAllColumns

100 ps

LayoutSimulate

Search:

documents/ModelSim/PROJECT/fulladder Gate level/Fulladder

Name	Status	Type	Order	M
multiplier_tb sv	✓	SystemVerilog	8	0t
ripple_carryadder sv	✓	SystemVerilog	3	0t
fulladder_gatelevel_tb sv	✓	SystemVerilog	2	0t
rcadder_16bit sv	✓	SystemVerilog	7	0t
multiplier_8x8_gatelevel...	✓	SystemVerilog	6	0t
ArithmeticLogicUnit_8bit.s...	✓	SystemVerilog	4	0t
ArithmeticLogicUnit_8bit_...	✓	SystemVerilog	5	0t
fulladder_gatelevel sv	✓	SystemVerilog	0	0t
halfadder_gatelevel sv	✓	SystemVerilog	1	0t

Objects

Name	Value	Kind	5 ps
a	0000...	Pack...	Internal
b	0000...	Pack...	Internal
sel	00	Pack...	Internal
result	0000...	Net	Internal

Processes (Hierarchy)

Name	Type (filtered)	State
#ALWAYS#42	Always	Wait
#INITIAL#15	Initial	Active

LibraryProjectsim

Transcript

Loading work.ArithmeticLogicUnit_8bit_tb
Loading work.ArithmeticLogicUnit_8bit
Loading work.rcadder_8bit
Loading work.fulladder_gatelevel
Loading work.halfadder_gatelevel
add wave sim:/ArithmeticLogicUnit_8bit_tb/*
add wave -r sim:/ArithmeticLogicUnit_8bit_tb/*
VSIM 23> run -all
Time=0 sel=00 a= 15 b= 10 result= X
Time=4 sel=00 a= 15 b= 10 result= X
Time=6 sel=00 a= 15 b= 10 result= 25
Time=20 sel=01 a= 15 b= 10 result= 5
Time=40 sel=10 a= 13 b= 5 result= 65
Time=60 sel=11 a= 20 b= 4 result= 5
** Note: \$finish : C:/Users/Ragavan/Downloads
/Documents/ModelSim/PROJECT/fulladder Gate level/A
rithmeticLogicUnit_8bit_tb sv(37)
Time: 80 ps Iteration: 0 Instance: /Arithme

M :/Users/Ragavan/Downloads/Documents/ModelSim/PROJECT/fulladder Gate level/ArithmeticLogicUnit_8bit_tb sv (/ArithmeticLogicUnit_8bit_tb) - Default

Ln#	
1	module ArithmeticLogicUnit_8bit_tb();
2	
3	reg [7:0] a;
4	reg [7:0] b;
5	reg [1:0] sel;
6	
7	wire [15:0] result;
8	ArithmeticLogicUnit_8bit DUT(
9	.a(a,
10	.b(b,
11	.sel(sel),
12	.result(result)
13	);
14	
15	initial
16	begin
17	a = 8'd15;
18	b = 8'd10;
19	sel = 2'b00;
20	#20;
21	
22	a = 8'd15;
23	b = 8'd10;
24	sel = 2'b01;
25	#20;
26	
27	a = 8'd13;
28	b = 8'd5;
29	sel = 2'b10;
30	#20;
31	
32	a = 8'd20;
33	b = 8'd4;
34	sel = 2'b11;
35	#20;
36	
37	\$finish;
38	end
39	
40	initial
41	begin
42	\$monitor("Time=%0t sel=%b a=%d b=%d result=%d",
43	\$time, sel, a, b, result);
44	end
45	endmodule
46	

rcadder_16bit sv

multiplier_8x8_gatelevel sv

ripple_carryadder sv

ArithmeticLogicUnit_8bit sv

ArithmeticLogicUnit_8bit_tb sv

Ln: 1 Col: 0

Project : Fulladder Now: 80 ps Delta: 0

sim:/ArithmeticLogicUnit_8bit_tb/#INITIAL#15

